1. O/N/23/21/Q1

Fig. 1.1 shows a toy water rocket.

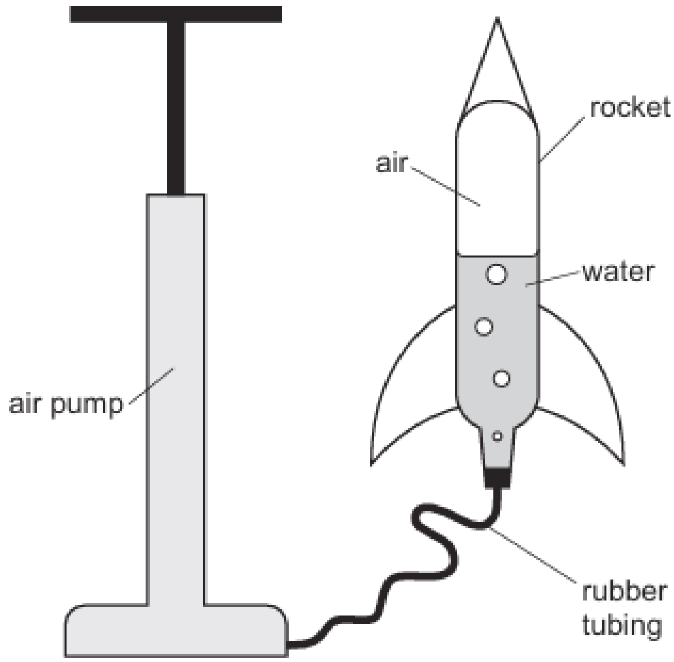

Fig. 1.1

The rocket is half-filled with water and connected by rubber tubing to a pump.

(a) Air is then pumped into the rocket so that the pressure of the air inside the rocket increases.

(i) Explain, in terms of particles, why the air in the rocket exerts a pressure on the walls of the rocket.

....

....

....

.... [3]

(ii) The temperature of the air in the rocket remains constant as air is pumped into the rocket.

Explain why increasing the number of air particles in the rocket increases the pressure of the air in the rocket.

....

.... [1]

2. O/N/23/22/Q4

Fig. 4.1 shows a long vertical tube, sealed at the base and open at the other end.

The tube is 1.0 m long.

The cross-sectional area of the tube is $4.0 \times 10^{-4}\,m^2$.

It is filled with a liquid of density $1.4 \times 10^4\,kg/m^3$.

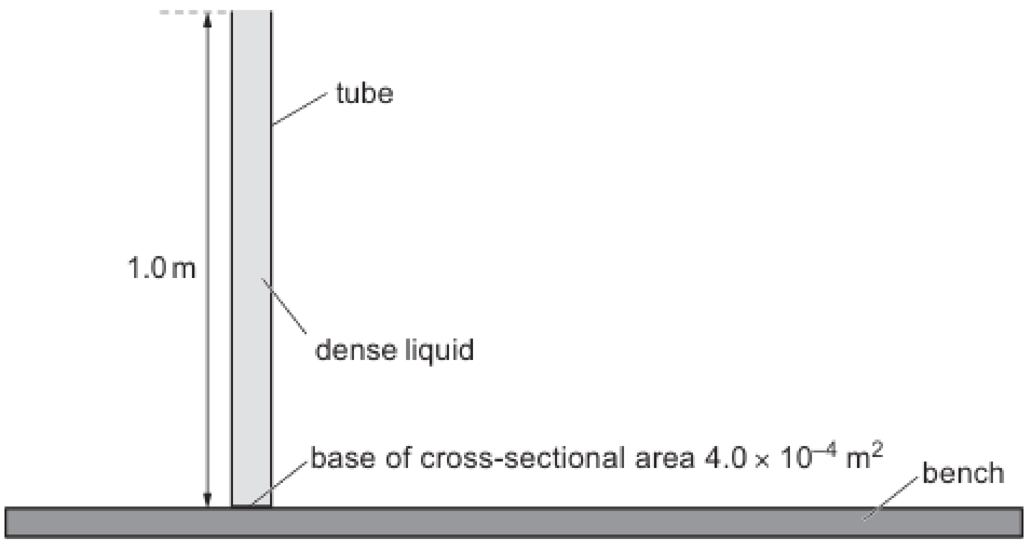

Fig. 4.1 (not to scale)

The atmospheric pressure is 1.0×10^5 Pa.

(a) Calculate:

(i) the total pressure in the liquid at the bottom of the tube

total pressure = ... Pa [3]

(ii) the force exerted on the inside surface of the bottom of the tube.

force = ... N [2]

(b) A small sheet of glass is placed over the open end of the tube.

The tube is inverted in a container of the dense liquid.

The open end of the tube and the sheet of glass are a short distance below the surface of the liquid.

Fig. 4.2 shows the arrangement.

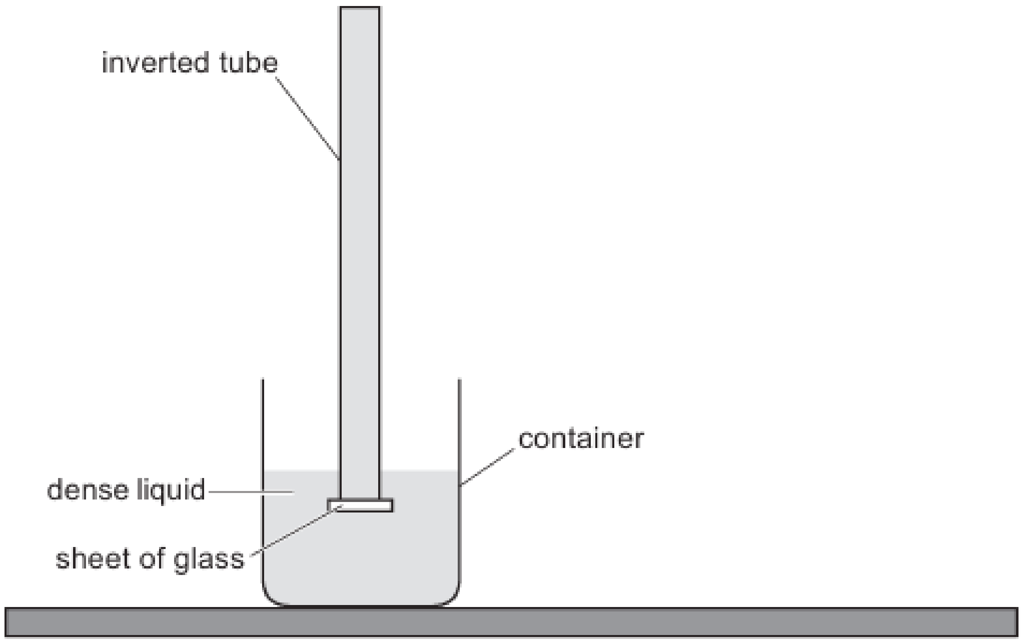

Fig. 4.2 (not to scale)

The sheet of glass seals the open end of the inverted tube.

(i) Describe what happens in the inverted tube when the sheet of glass is removed.

....

....

....

....

.... [3]

(ii) This equipment is used to make a measurement that is used to calculate atmospheric pressure.

Describe the measurement made and then used in the calculation of atmospheric pressure.

You may draw a diagram to help your description.

....

....

.... [2]

[Total: 10]

3. M/J/23/21/Q4

(a) A student pushes a drawing pin into a wooden board, as shown in Fig. 4.1.

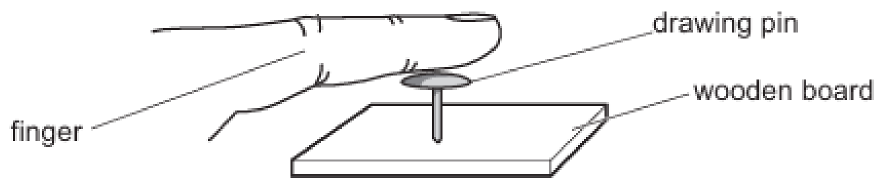

Fig. 4.1

The area of the pin in contact with the finger is $5.0 \times 10^{-5}\,m^2$. The student pushes downwards with a force of 26 N.

The mass of the drawing pin is very small.

(i) Calculate the pressure exerted by the finger on the drawing pin.

pressure = ... Pa [2]

(ii) Compare the force exerted by the finger on the drawing pin with the force exerted by the drawing pin on the wooden board.

....

.... [1]

(iii) Explain why the drawing pin goes into the wooden board but not into the finger.

....

....

....

.... [2]

(b) Fig. 4.2 shows water emerging from a plastic bag that contains a number of small holes.

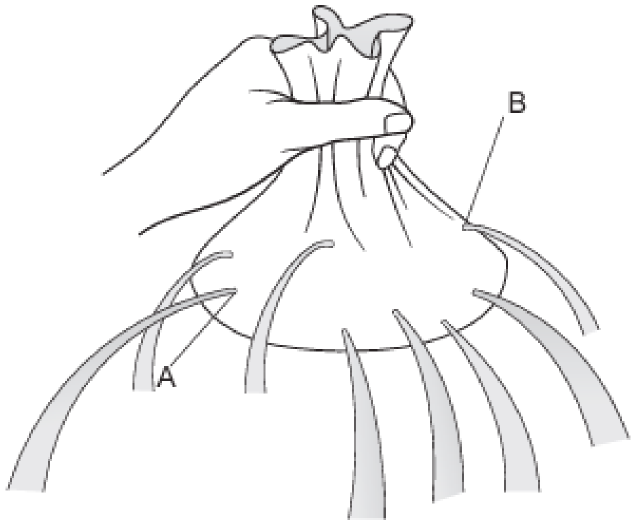

Fig. 4.2

(i) Explain why the water emerges from each hole in a direction at right angles to the surface of the bag.

....

....

....

.... [2]

(ii) The holes at A and B are the same size.

Explain why the water emerges faster from the hole at A than from the hole at B.

....

....

.... [1]

[Total: 8]

4. M/J/23/22/Q3

A fixed mass of gas in a glass tube is trapped by a seal at one end of the tube and by a column of mercury. The mercury is free to move within the tube.

The tube is rotated slowly from the vertical as shown in Fig. 3.1 to the horizontal as shown in Fig. 3.2. The volume of the gas increases and its temperature remains constant.

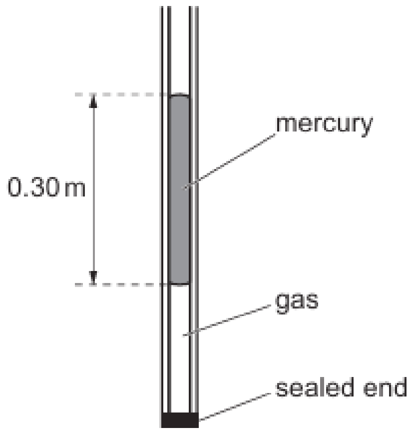

Fig. 3.1 (not to scale)

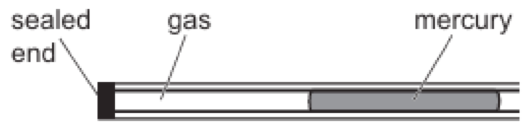

Fig. 3.2 (not to scale)

(a) (i) Describe why rotating the tube changes the pressure of the gas in the sealed end.

....

....

.... [1]

(ii) Explain, using ideas about particles, why the pressure of the gas decreases when its volume increases.

....

....

....

.... [3]

(b) In Fig. 3.1 the length of the mercury column is 0.30 m.

The density of mercury is 14 000 kg / m^3.

Atmospheric pressure is 1.0×10^5 Pa.

Calculate the pressure of the gas in the tube.

pressure = ... Pa [3]

(c) The pressure of a different sample of gas changes at constant temperature.

Fig. 3.3 shows one point, marked X, on a graph of pressure against volume for the gas sample.

At X the pressure of the gas is P_0 and its volume is V_0.

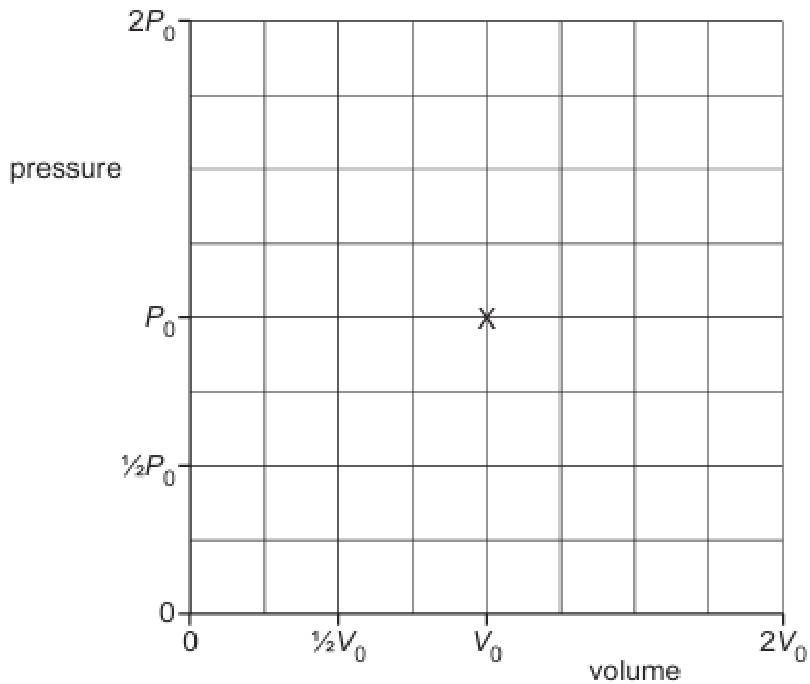

Fig. 3.3

On Fig. 3.3, sketch the graph as the pressure of the gas decreases from $2P_0$ to $\frac{1}{2}P_0$. [2]

[Total: 9]

5. M/J/22/22/Q3

Fig. 3.1 shows a syringe mounted vertically in a block of wood and sealed at one end. A plunger is free to move inside the syringe.

There is trapped air in the syringe.

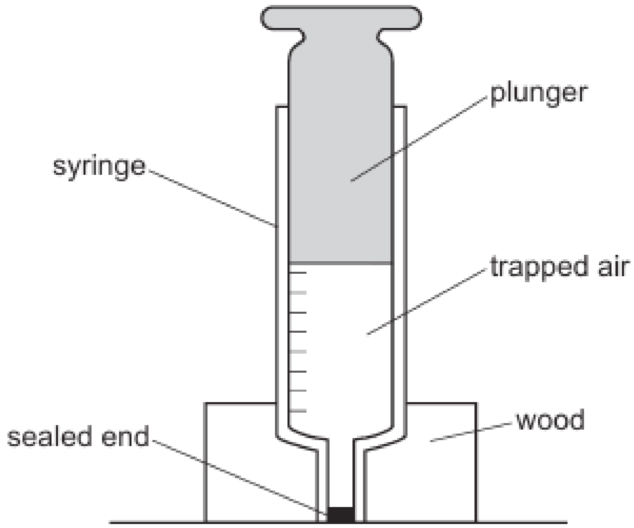

Fig. 3.1

The air inside the syringe exerts a pressure on the walls of the syringe.

(a) Define the term pressure.

....

....

.... [1]

(b) Explain how the air molecules in the cylinder of the syringe create a pressure.

....

....

....

....

.... [3]

(c) A 10 N weight is placed on top of the plunger. The plunger moves down slowly so that the temperature of the air inside the syringe does not change.

Before the weight is placed on top of the plunger:

- the pressure of the air inside the syringe is 1.0×10^5 Pa
- the volume of the air is 50 cm^3.

The cross-sectional area of the plunger is 1.2×10^{-4} m^2.

(i) Calculate the pressure of the air in the syringe after the plunger stops moving.

pressure = ... [2]

(ii) Calculate the volume of air inside the syringe after the plunger stops moving.

volume = ... [2]

[Total: 8]

6. O/N/20/21/Q8

The density of water in a lake is $1000\,kg/m^3$.

At a depth of 25 m beneath the surface of the lake, the total pressure is $3.5 \times 10^5\,Pa$.

(a) State what is meant by *pressure*.

....

.... [1]

(b) The gravitational field strength is 10 N/kg.

Determine:

(i) the pressure due to 25 m of water

pressure = ... [2]

(ii) the atmospheric pressure.

atmospheric pressure = ... [1]

(c) An underwater depth gauge contains a small cylinder as shown in Fig. 8.1. Gas is trapped inside the cylinder by a piston. The piston is free to move.

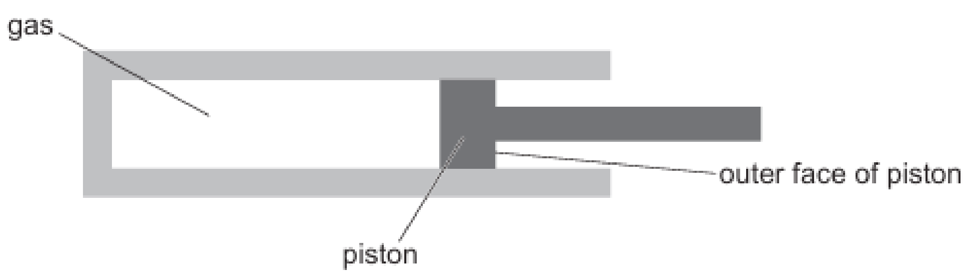

Fig. 8.1

The outer face of the piston is in contact with the water.

As the depth gauge is lowered into the water, the piston moves into the cylinder. This moves a needle on a dial to indicate the depth of the gauge in the water.

(i) Explain why the piston moves into the cylinder.

....

....

....

....

.... [3]

(ii) The temperature of the gas does not change as the piston moves into the cylinder.

Explain, in terms of molecules, what happens to the pressure of the trapped gas as the piston moves into the cylinder.

....

....

....

.... [3]

(iii) At the surface of the water, the volume of the trapped gas in the depth gauge is V_0.

On Fig. 8.2, sketch a graph to show how the volume of trapped gas decreases as the gauge is lowered into the water.

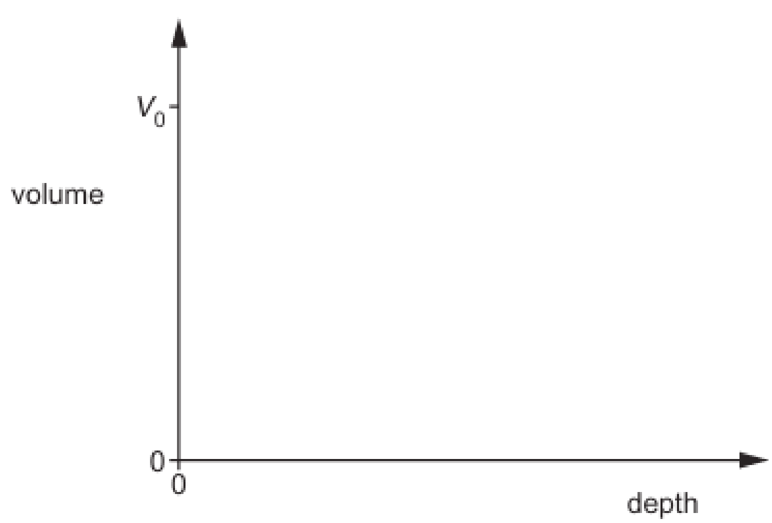

Fig. 8.2

[2]

(iv) The instructions for the depth gauge state that, each time it is used, the needle of the dial must be re-set to zero at the surface of the water.

Suggest **one** reason for this.

....

.... [1]

(v) The density of the air trapped in the depth gauge increases. The density of the water remains constant.

Explain, in terms of the molecules of the water, why the density of the water remains constant.

....

....

.... [2]

[Total: 15]

7. O/N/19/21/Q1

Fig. 1.1 shows a tube full of mercury placed upside down in a dish of mercury. Initially a small piece of glass closes the open end of the tube.

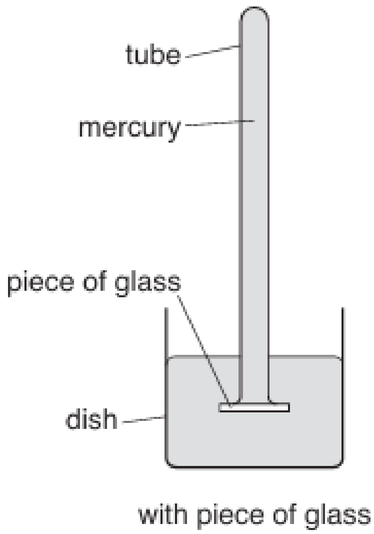

Fig. 1.1

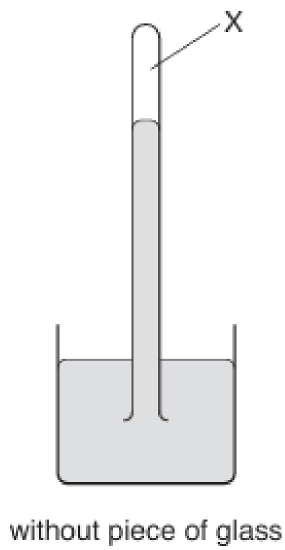

Fig. 1.2

(a) The piece of glass is removed and the level of mercury in the inverted tube falls, as shown in Fig. 1.2.

(i) Explain why the level of mercury in the tube falls.

....

....

....

.... [2]

(ii) State what is found at point X in Fig. 1.2.

.... [1]

(b) Describe how the tube of mercury in Fig. 1.2 is used to determine a value for the atmospheric pressure in pascals (Pa). You may draw on Fig. 1.2 if you wish.

....

....

....

.... [3]

[Total: 6]

8. O/N/19/22/Q2

A student uses a pump to inflate a bicycle tyre.

Fig. 2.1 shows the pump and the tyre.

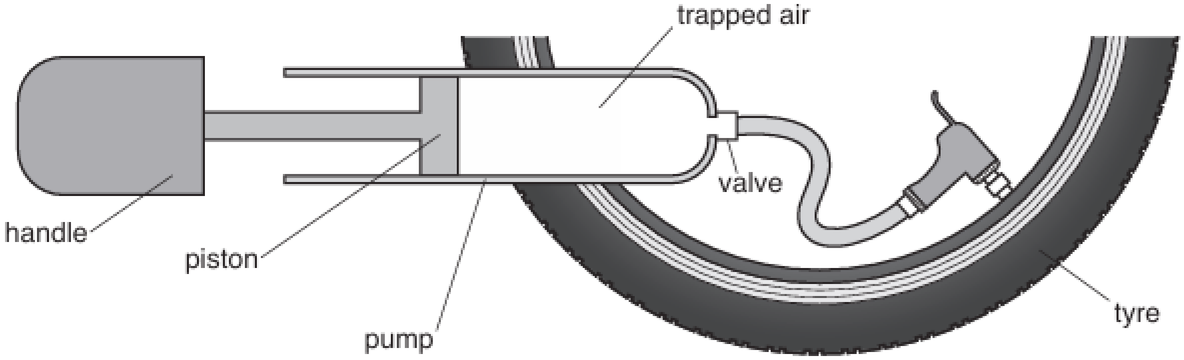

Fig. 2.1 (not to scale)

(a) The pressure of the trapped air in the pump is 3.8×10^5 Pa and the cross-sectional area of the piston is 6.1×10^{-4} m^2.

(i) Calculate the force exerted on the piston by the trapped air in the pump.

force = ... [2]

(ii) The student pushes the handle to the right and the piston forces the trapped air into the tyre. The force exerted by the student is less than the value in **(a)(i)**.

Suggest one reason why.

....

.... [1]

(b) The temperature of the air in the pump remains constant as the handle moves to the right.

Explain, in terms of its molecules, why the pressure of a gas increases when its volume is decreased at constant temperature.

....

....

....

....

.... [3]

[Total: 6]

9. O/N/18/21/Q2

A small crack appears in an oil tank and a stream of oil is pushed out through the crack. The oil hits the floor at P where a puddle of oil starts to form.

Fig. 2.1 shows the oil tank.

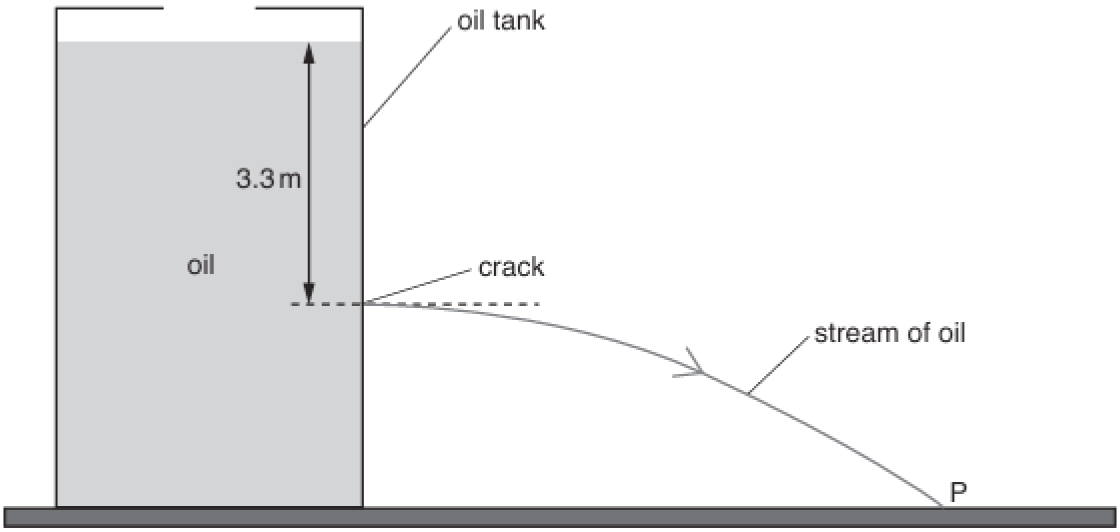

Fig. 2.1

The gravitational field strength g is equal to $10\,N/kg$.

(a) The density of the oil is $940\,kg/m^3$ and the crack is 3.3 m below the surface of the oil.

(i) Calculate the pressure due to the oil at the level of the crack.

pressure = ... [2]

(ii) Explain why the atmospheric pressure does not affect the rate at which the oil is pushed out through the crack.

....

.... [1]

(b) As time passes, the point where the oil hits the floor moves away from P and towards the tank.

Explain why this happens.

....

....

.... [2]

10. O/N/17/21/Q2

Fig. 2.1 shows a can of compressed air that is being used to blow dust off a computer keyboard.

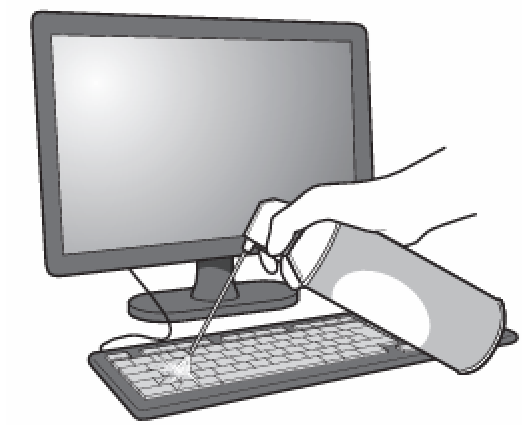

Fig. 2.1

(a) The pressure of the air inside the can is greater than the pressure of the atmosphere.

(i) State what is meant by the term *pressure*.

....

.... [1]

(ii) Explain, in terms of molecules, why the pressure of the air inside the can decreases as it is used.

....

....

....

....

....

.... [3]

11. O/N/15/22/Q3

A very deep tank is used when training sailors to escape from submarines. The tank is cylindrical and open to the air at the top. Fig. 3.1 shows the tank.

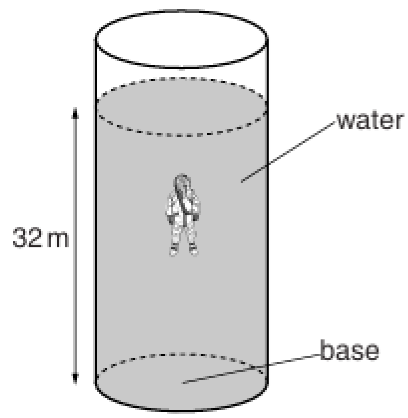

Fig. 3.1 (not to scale)

The area of the base of the tank is $45\,m^2$ and the tank is filled with water to a depth of 32 m.

(a) The density of water is $1000\,kg/m^3$ and the gravitational field strength g is 10 N/kg.

Calculate the pressure, due to the water, on the base of the tank.

pressure = ... [2]

(b) The pressure in the water at the base of the tank is 4.2×10^5 Pa.

(i) Explain why the pressure in the water at the base of the tank differs from the value calculated in **(a)**.

....

....

.... [1]

(ii) Calculate the force exerted on the base of the tank.

force = ... [2]

12. M/J/15/22/Q2

Fig. 2.1 shows two engineers measuring the length of a wall made from concrete.

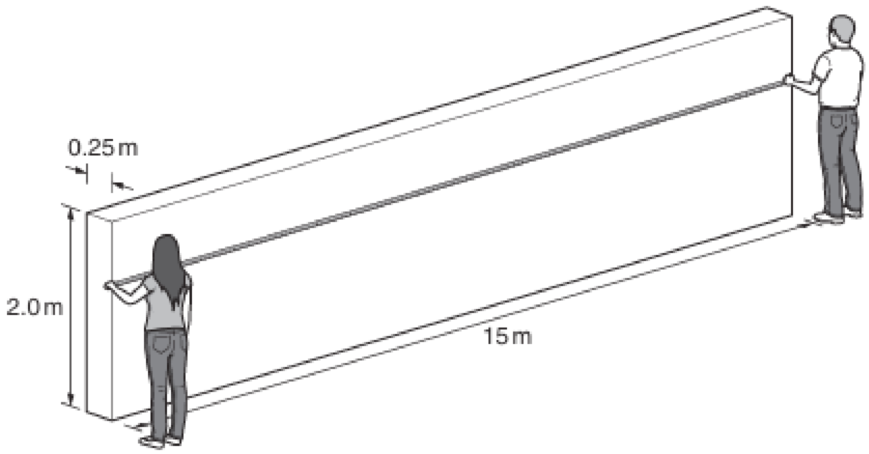

Fig. 2.1 (not to scale)

The wall is 2.0 m high, 15 m long and 0.25 m thick.

The weight of the wall is 180 000 N and the mass of the wall is 18 000 kg.

(a) The engineers measure the length of the wall in one single measurement.

State the name of the measuring instrument they use.

.... [1]

(b) The engineers state that the density of the concrete affects the pressure exerted by the wall on the ground but that the length of the wall does not affect this pressure.

(i) Define *density*.

....

.... [1]

(ii) Calculate the density of the concrete.

density = ... [2]

(iii) Calculate the pressure exerted by the wall on the ground.

pressure = ... [2]

(iv) Without further calculation, explain why doubling the length of the wall does not change the pressure that the wall exerts on the ground.

....

....

....

.... [1]

13. M/J/14/21/Q9

A garden pond contains a small fountain. An electric pump in the water causes the water to rise above the surface of the pond, as shown in Fig. 9.1.

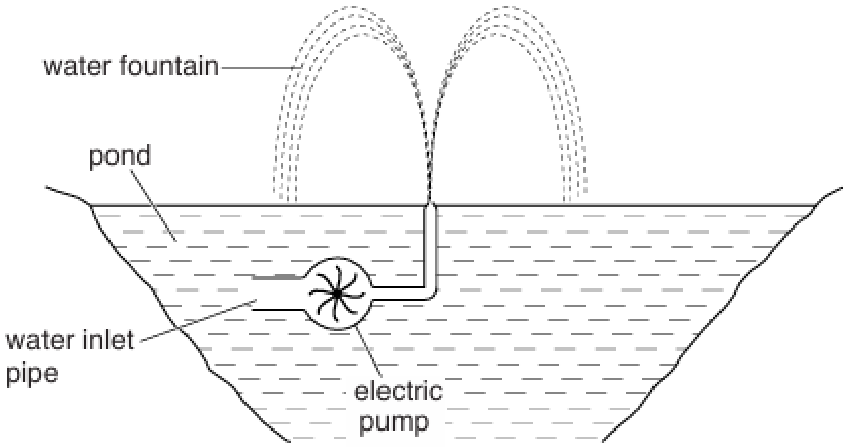

Fig. 9.1

(a) The pressure of the water in the pond increases with depth.

(i) Explain what is meant by *pressure*.

....

.... [1]

(ii) Explain why the pressure below the surface of the water increases with depth.

....

.... [2]

(iii) State the unit of pressure.

.... [1]

(b) Describe the energy changes that occur within the pump.

....

....

.... [3]

14. M/J/13/21/Q3

Fig. 3.1 shows a glass tube dipped into mercury. A vacuum pump is connected to the top of the tube and switched on. The mercury rises up the tube and stops.

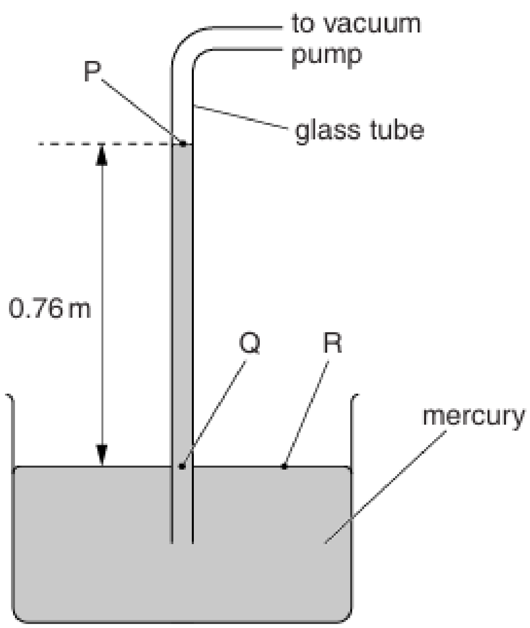

Fig. 3.1

(a) Three points on Fig. 3.1 are labelled P, Q and R.

State which **two** of these points are at atmospheric pressure.

.... [1]

(b) The density of mercury is 13 600 kg / m^3 and the gravitational field strength g is 10 N / kg.

Calculate the pressure due to the column of mercury of length 0.76 m.

pressure = ... [2]

(c) State and explain what happens if the mercury in the apparatus shown in Fig. 3.1 is replaced with water.

....

....

.... [2]

15. M/J/12/21/Q3

A student measures the pressure inside a bicycle tyre using a pressure gauge he has constructed. Fig. 3.1 shows the apparatus he uses. The piston and rod move along the smooth cylindrical tube.

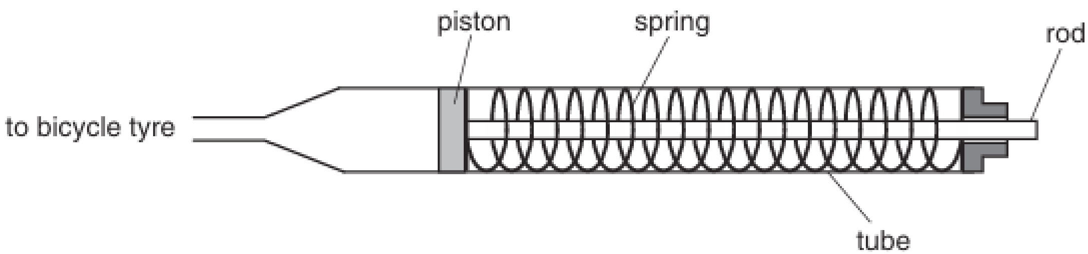

Fig. 3.1

The change in length of the spring is proportional to the force applied to the spring. A force of 2.8 N compresses the spring by 2.0 cm.

The student connects the pressure gauge to the tyre. Air from the tyre exerts a force on the piston. As the piston moves, the spring is compressed. The piston moves 10 cm to the right and then stops.

(a) Calculate the force exerted on the piston by the spring.

force = ...[1]

(b) The cross-sectional area of the piston is $3.0 \times 10^{-5}\,m^2$. Calculate the pressure of the air inside the tyre.

pressure = ...[2]

(c) Suggest one change to the apparatus so that the piston moves a smaller distance when measuring the same pressure.

....[1]

(d) Each time that the pressure gauge is used, the pressure in the tyre falls slightly. The temperature stays constant. Describe, using ideas about molecules, why the pressure falls.

....

....

....

....[2]

16. O/N/09/Q1

A diver holds his breath and dives into the sea from a boat to a depth of 25.0 m. The atmospheric pressure is 1.05×10^5 Pa.

(a) **(i)** Explain why the pressure at this depth is greater than the atmospheric pressure.

....

....[1]

(ii) Other than the depth and the atmospheric pressure, state **one** quantity that affects the pressure in a liquid.

....

....[1]

(b) **(i)** The pressure due to 25.0 m of sea-water is 2.55×10^5 Pa. Calculate the total pressure at this depth.

pressure = ...[1]

(ii) As the diver holds his breath and descends to a depth of 25.0 m, the greater pressure causes the volume of the air trapped in his lungs to change.

When he is on the boat, the total volume of the air in his lungs is 6000 cm^3. Calculate the volume of this air at a depth of 25.0 m.

volume = ...[2]